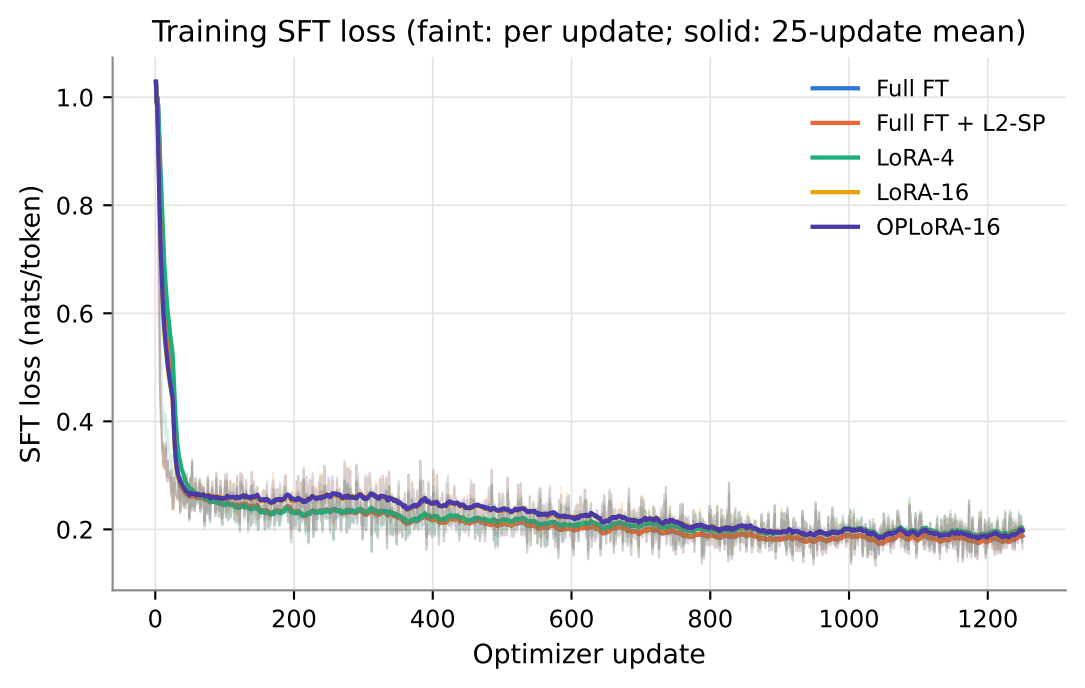

Training SFT loss (faint: per update; solid: 25-update mean)
SFT loss (nats/token)
1.0
0.8
0.6
0.4
0.2
0
200
400
600
800
1000
1200
Optimizer update
Full FT
Full FT + L2-SP
LoRA-4
LoRA-16
OPLoRA-16